



B. Tech – II

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Unit-3

Laplace Transform

Topics in the Module

- Laplace transforms, Advantages and sufficient conditions for Existence of Laplace transform
- Laplace transform of standard functions
- General properties of Laplace transforms and problems based on it.
- Laplace transform of periodic function: Statement and problems.
- Laplace transform of Unit step function
- Second shifting property
- Laplace transform of unit impulse function

CLASS-1

INTRODUCTION TO LAPLACE TRANSFORM

LAPLACE TRANSFORM

- TRANSFORM implies a major change in form, nature, or function.
- **Transform** in mathematics means a mathematical function from one domain to other or on to itself
- The Laplace transform is named after mathematician and astronomer [Pierre-Simon Laplace](#).
- Laplace transform is an [integral transform](#) perhaps second only to the [Fourier transform](#) in its utility in solving physical problems. The Laplace transform is particularly useful in solving linear [ordinary differential equations](#) such as those arising in the analysis of electronic circuits.

Definition of Integral Transform

A Laplace transform is a type of integral transform of a function $f(t)$ from time domain to the complex frequency domain $F(s)$.

An **integral transform** of a function f is a relation of the form

$$F(s) = \int_{\alpha}^{\beta} K(s, t) f(t) dt, \quad \infty \leq \alpha < \beta \leq \infty$$

Given a known function $K(s, t)$, called kernel function

Plug one function in $f(t)$

Get another function out $F(s)$

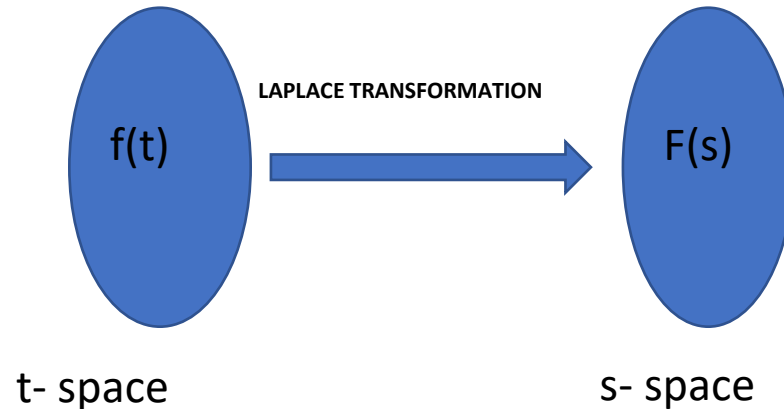
The new function is in a different domain

Definition of Laplace

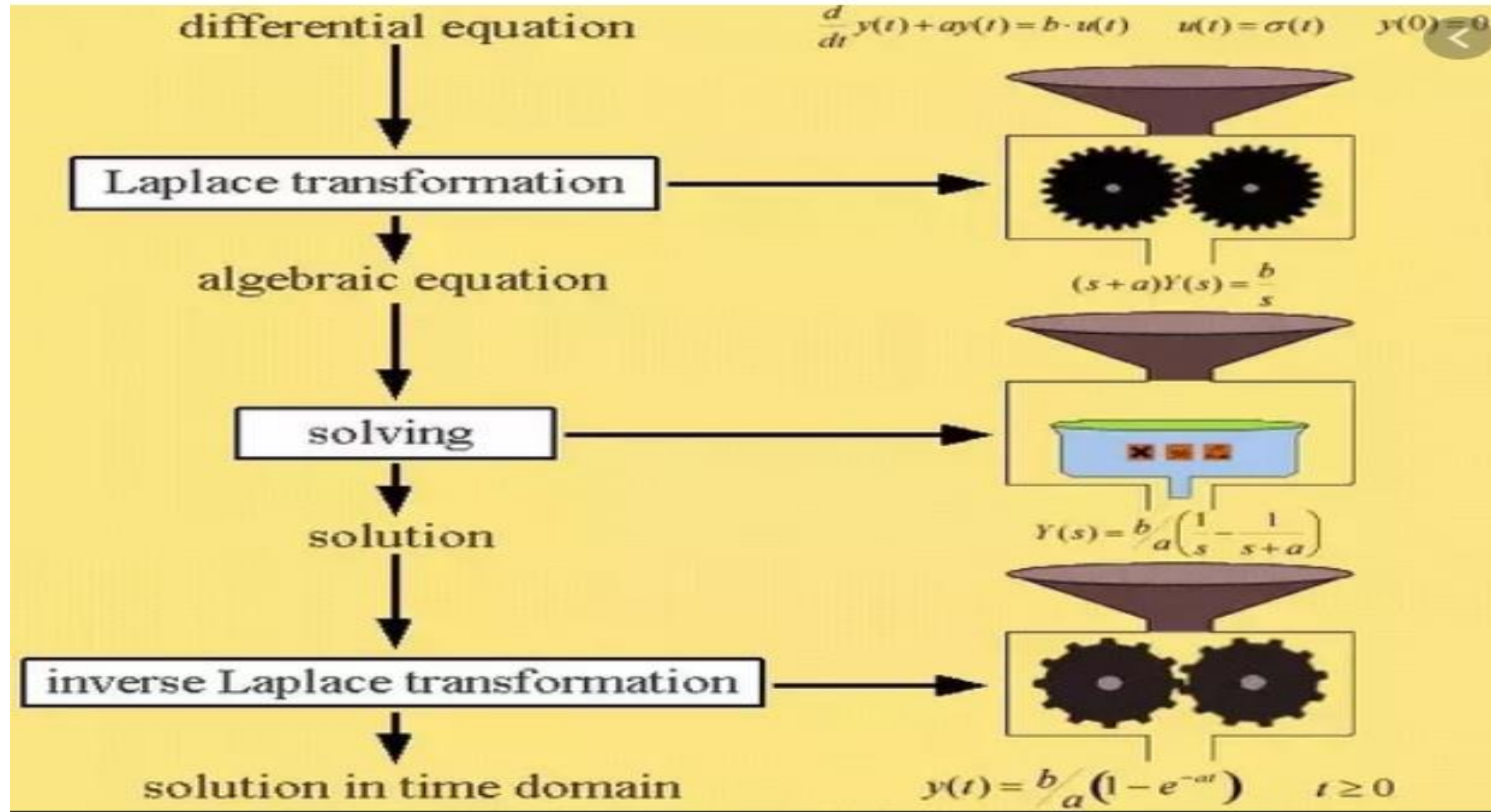
- Let $f(t)$ be defined for $t \geq 0$ and let s be a real /complexnumber. Then the *Laplace transform* of $f(t)$ is the function $F(s)$ defined by

$$F(s) = \int_0^{\infty} e^{-st} f(t) dt = L\{f(t)\}$$

- for those values of s for which the improper integral converges.



What does Laplace transformation do?



Laplace Transform

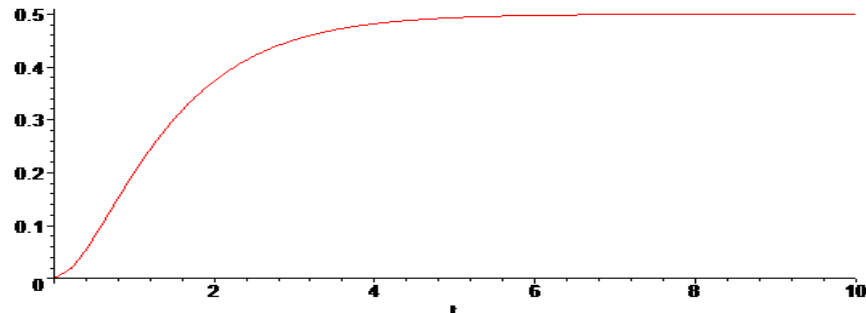
- Laplace transforms

Time Domain

$$y^{(2)}(t) + y^{(1)}(t) + y(t) = x(t)$$

$$x(t) = 1$$

$$y(t) = \frac{1}{2} - e^{-t} + \frac{1}{2} e^{-2t}$$



Laplace
transform

Frequency Domain

$$H(s) = \frac{1}{s^2 + 3s + 2}$$

$$X(s) = \frac{1}{s}$$

Solve algebraic equation

$$\frac{1}{s} \frac{1}{s^2 + 3s + 2}$$

Inverse Laplace
transform

Existence of Laplace Transforms

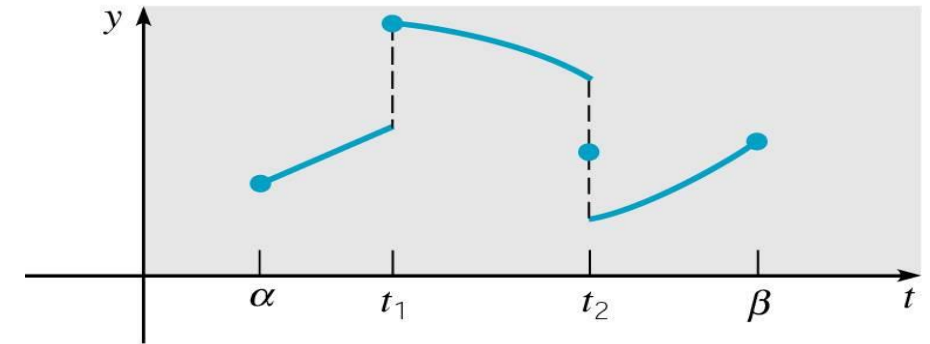
- Do every function has a Laplace transform?
- $\int_0^{\infty} e^{-st} e^{t^2} dt = \infty$ for very real number s . Hence, for the function $f(t) = e^{t^2}$ does not have a Laplace transform.
- Our next objective is to establish conditions that ensure the existence of the Laplace transform of a function

Piecewise Continuous Functions

- A function f is **piecewise continuous** on an interval $[a, b]$ if this interval can be partitioned by a finite number of points

$a = t_0 < t_1 < \dots < t_n = b$ such that

- (1) f is continuous on each (t_k, t_{k+1})
- (2) $\left| \lim_{t \rightarrow t_k^+} f(t) \right| < \infty, \quad k = 0, \dots, n-1$
- (3) $\left| \lim_{t \rightarrow t_{k+1}^-} f(t) \right| < \infty, \quad k = 1, \dots, n$



- In other words, f is piecewise continuous on $[a, b]$ if it is continuous there except for a finite number of jump discontinuities.

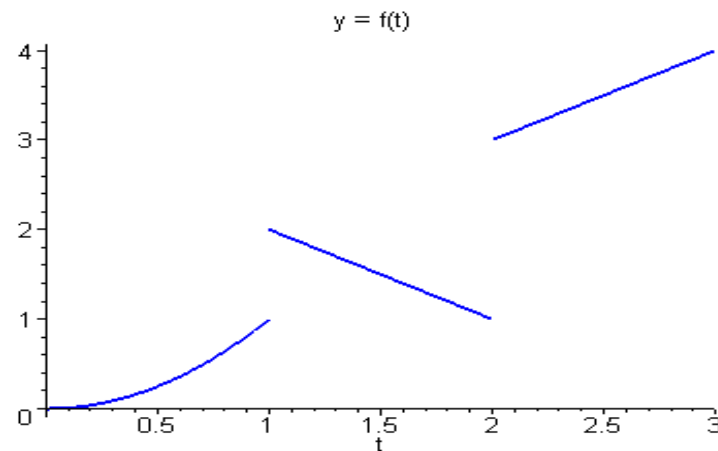
Note that a piecewise continuous function is a function that has a finite number of breaks in it and doesn't blow up to infinity anywhere

Example

- Consider the following piecewise-defined function f .

$$f(t) = \begin{cases} t^2, & 0 \leq t \leq 1 \\ 3 - t, & 1 < t \leq 2 \\ t + 1 & 2 < t \leq 3 \end{cases}$$

- From this definition of f , and from the graph of f below, we see that f is piecewise continuous on $[0, 3]$.



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$$\int_0^{\infty} e^{-st} e^{t^2} dt = \lim_{T \rightarrow \infty} \int_0^T e^{-st} e^{t^2} dt$$

- do piecewise continuity alone does not guarantee that the improper integral converges
- $\int_0^{\infty} e^{-st} e^{t^2} dt = \infty$. this occurs because e^{t^2} increases too rapidly as $t \rightarrow \infty$. The next definition provides a constraint on the growth of a function e^{t^2} that guarantees convergence of its Laplace transform for s in some interval (s_0, ∞) .

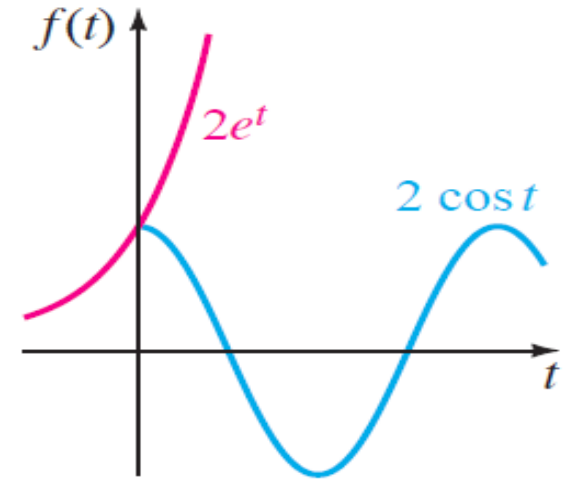
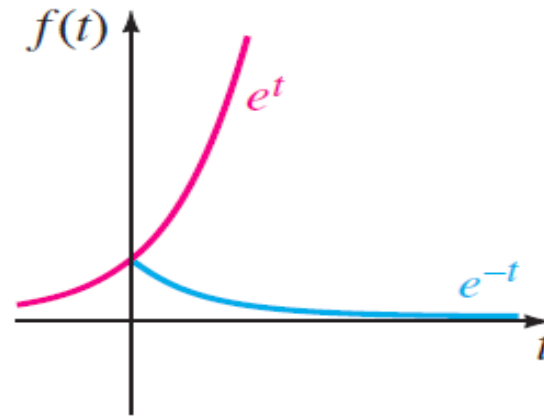
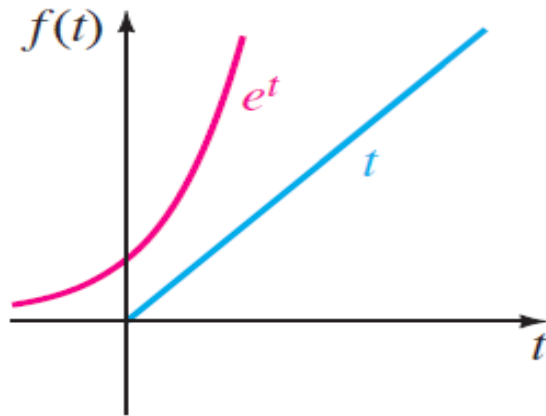
Exponential order

- A function f is said to be *of exponential order* s_0 if there are constants M and t_0 such that
- $|f(t)| \leq Me^{s_0 t}$, $t > t_0$
- In situations where the specific value of s_0 is irrelevant we say simply that f is *of exponential order*.

Condition for Laplace transform to exists

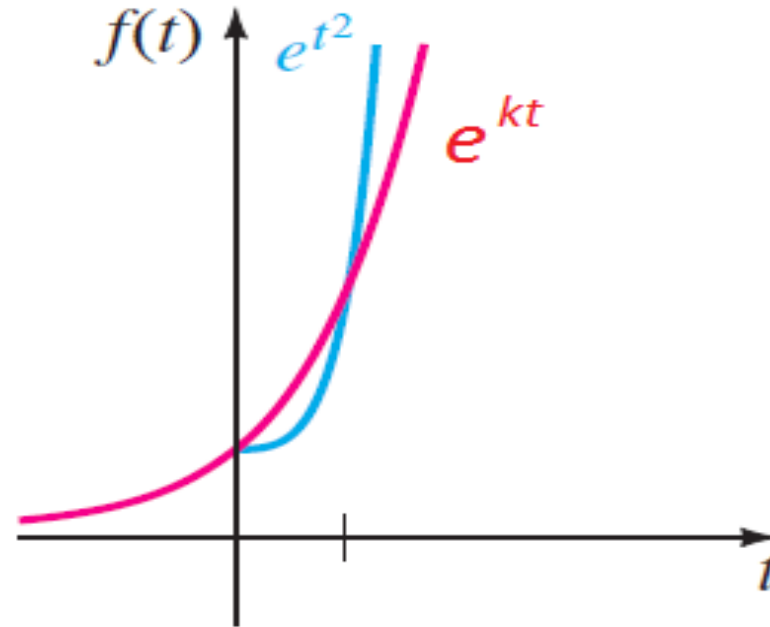
- If f is piecewise continuous on $[0, \infty)$ and of exponential order s_0 , then $L(f)$ is defined for $s > s_0$
- The above theorem gives a **sufficient condition** for the existence of Laplace transforms.

Exponential order(Bounded)...



Functions with blue graphs are of exponential order

Exponential order(Bounded)...



$f(t) = e^{t^2}$ is not of exponential order since its graph grows faster

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Thank You